Lecture 6: Parallel programming

Threads, channels and shared state

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DevLab Rust 2025

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github.com/wvhulle/rust-course-ghent

Make sure you have already finished:

- session-5/tests/error-handling.rs

1.1. Plain Threads

```rust
use std::thread;
use std::time::Duration;

fn main() {
    thread::spawn(|| {
        for i in 0..10 {
            println!("Count in thread: {i}!");
            thread::sleep(Duration::from_millis(5));
        }
    });

    for i in 0..5 {
        println!("Main thread: {i}");
        thread::sleep(Duration::from_millis(5));
    }
}
```

(playground link)

What happens to the spawned thread?

1.1. Plain Threads

```rust
use std::thread;
use std::time::Duration;

fn main() {
    thread::spawn(|| {
        for i in 0..10 {
            println!("Count in thread: {i}!");
            thread::sleep(Duration::from_millis(5));
        }
    });

    for i in 0..5 {
        println!("Main thread: {i}");
        thread::sleep(Duration::from_millis(5));
    }
}
```

(playground link)

What happens to the spawned thread?

Terminated by operating system when main thread exits. No destructors called.

1.1. Plain Threads

```rust
thread::spawn(|| {
    for i in 0..10 {   }
});

for i in 0..5 {
    println!("Main thread: {i}");
    thread::sleep(Duration::from_millis(5));
}
```

(playground link)

How can we ensure the spawned thread finishes?

1.1. Plain Threads

```
1 thread::spawn(|| {
2     for i in 0..10 {   }
3 });
4
5 for i in 0..5 {
6     println!("Main thread: {i}");
7     thread::sleep(Duration::from_millis(5));
8 }                                                    (playground link)
```

How can we ensure the spawned thread finishes?

Use `JoinHandle` returned by `thread::spawn` and call `join()` on it.

```
1 fn main() {
2   let handle = thread::spawn(|| {
3     // thread code
4   });
5   // main thread code
6   handle.join();
7 }                                                    (playground link)
```

1.2. Receiving data from threads

How can we get a return value from a thread?

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How can we get a return value from a thread?

The `JoinHandle` returns a `Result` with the thread's return value.

```rust
fn main() {
    let handle = thread::spawn(|| {
        42
    });

    let result = handle.join().unwrap();
    println!("The answer is: {}", result);
}
```

(playground link)

What happens to the main thread if a spawned thread panics?

1.2. Receiving data from threads

How can we get a return value from a thread?

The `JoinHandle` returns a `Result` with the thread's return value.

```rust
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    let handle = thread::spawn(|| {
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    });

    let result = handle.join().unwrap();
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(playground link)

What happens to the main thread if a spawned thread panics?

Nothing, the main thread continues executing.

1.3. Handling thread panics

If we don't join, we never see the panic.

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If we don't handle the `Result` from `join()`, the panic is also ignored.

We need to check the `Result` from `join()` to see if the thread panicked:

```
1  fn main() {
2      let handle = thread::spawn(|| {
3          panic!("Thread panicked!");
4      });
5      let result = handle.join();
6      match result {
7          Ok(_) => println!("Thread completed successfully."),
8          Err(e) => println!("Thread panicked: {:?}", e),
9      }
10 }
```

(playground link)

Warning

You are responsible for handling panics in spawned threads! Rust will not do this for you.

Exercise:

- examples/join-handle.rs

1.4. Sending references to threads

Not all references can be sent to threads.

```rust
fn main() {
    let data = vec![1, 2, 3];
    let handle = thread::spawn(|| {
        let sum: i32 = data.iter().sum();
        println!("Sum: {sum}");
    });
    data.push(4); // modify while thread may be reading
    handle.join().unwrap();
}
```

(playground link)

What happens if a thread borrows data from the main thread?

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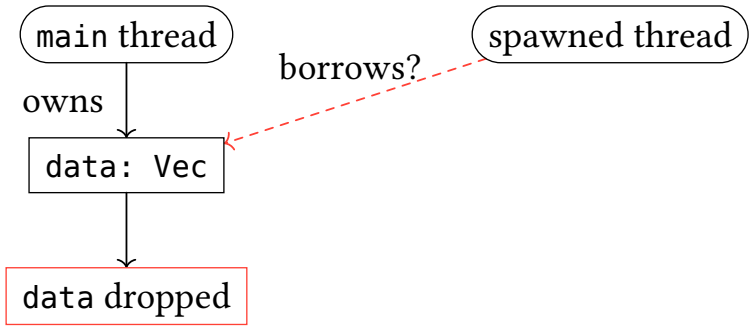
(playground link)

What happens if a thread borrows data from the main thread?

Compilation error: closure may outlive the current function, but it borrows `data`.



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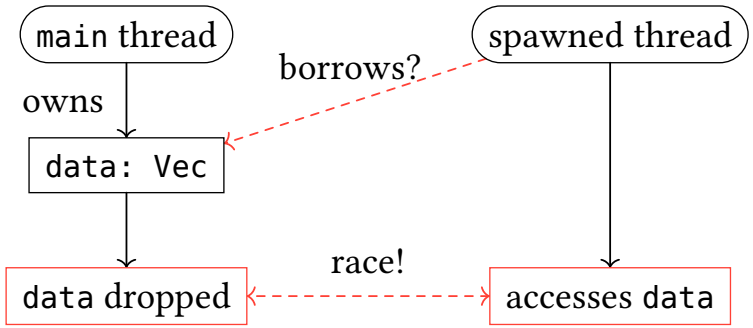

1.5. Borrowing and threads (continued)

First attempt: ensure the thread finishes before `main` exits.

```rust
fn main() {
    let data = vec![1, 2, 3];
    let handle = thread::spawn(|| {
        let sum: i32 = data.iter().sum();
        println!("Sum: {sum}");
    });
    handle.join().unwrap();
    println!("Original: {:?}", data); // try to use after thread
}
```

(playground link)

Does adding `join()` fix the borrow error?

1.5. Borrowing and threads (continued)

First attempt: ensure the thread finishes before `main` exits.

```rust
fn main() {
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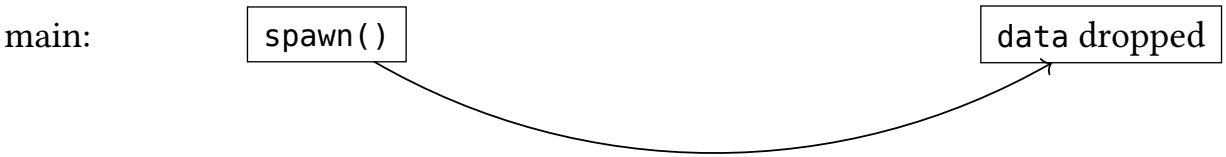
(playground link)

Does adding `join()` fix the borrow error?

No. Rust cannot verify at compile time that `join` is called before `data` goes out of scope.



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6     });
7     handle.join().unwrap();
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```

(playground link)

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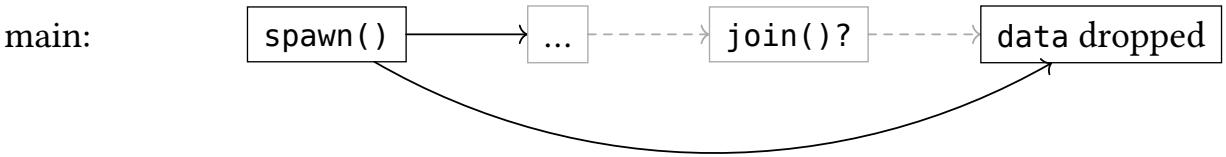

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    handle.join().unwrap();
    println!("Original: {:?}", data); // try to use after thread
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(playground link)

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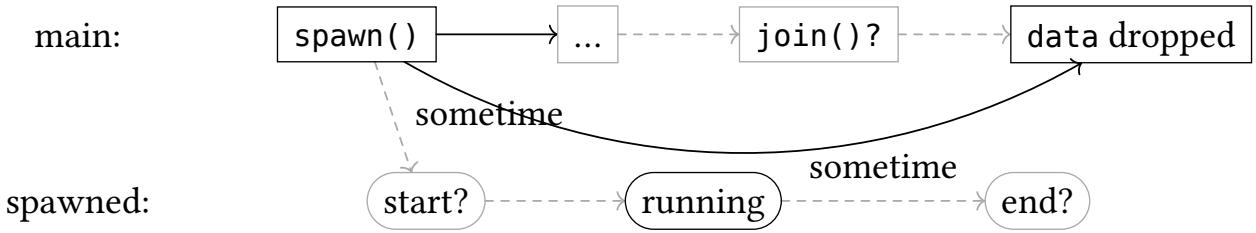

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First attempt: ensure the thread finishes before `main` exits.

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(playground link)

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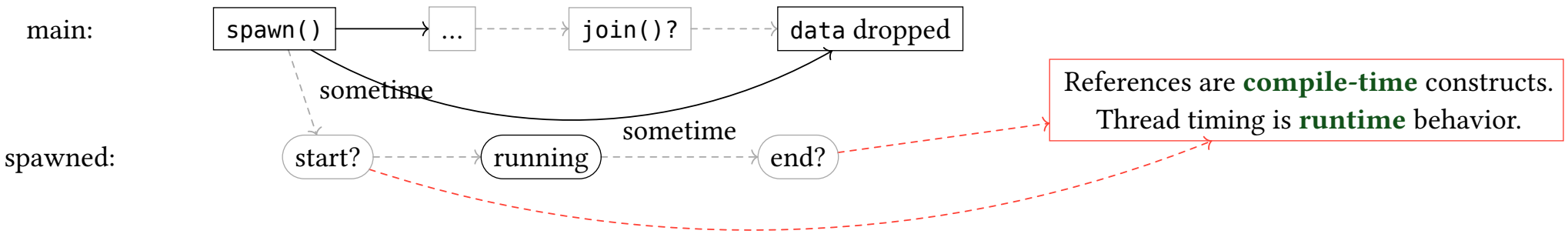

1.6. Scoped threads

The problem: `thread::spawn` requires `'static` because the thread may outlive the caller.

```
fn main() {
    let data = vec![1, 2, 3];
    let handle = thread::spawn(|| {
        println!("Data: {:?}", data); // error: `data` does not live long enough
    });
    handle.join().unwrap();
}
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(playground link)

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    let data = vec![1, 2, 3];
    let handle = thread::spawn(|| {
        println!("Data: {:?}", data); // error: `data` does not live long enough
    });
    handle.join().unwrap();
}
```

(playground link)

Solution: `thread::scope` guarantees all spawned threads complete before the scope ends.

```
fn main() {
    let data = vec![1, 2, 3];
    thread::scope(|s| {
        s.spawn(|| println!("Data: {:?}", data)); // borrowing works!
    }); // all threads joined here automatically
    println!("Back in main: {:?}", data);
}
```

(playground link)

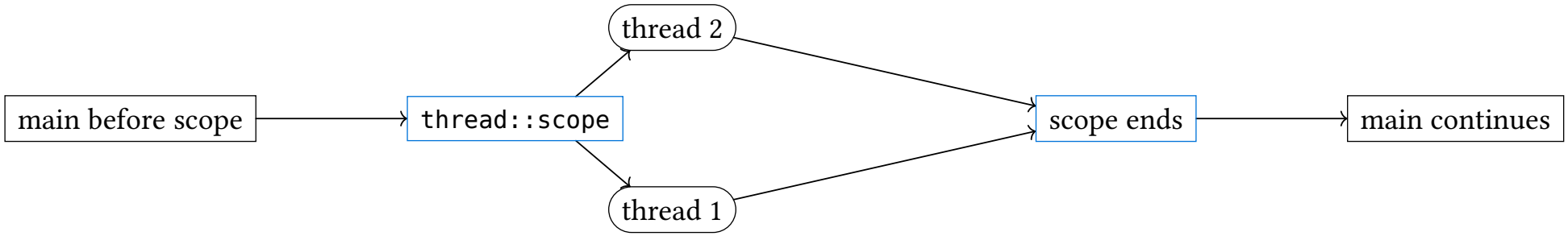
1.7. Multiple scoped threads

```rust
fn main() {
    let mut data = vec![1, 2, 3];

    thread::scope(|s| {
        s.spawn(|| println!("Thread 1: {:?}", data)); // shared borrow
        s.spawn(|| println!("Thread 2: {:?}", data)); // shared borrow
    });

    data.push(4); // mutable access after scope ends
}
```

(playground link)



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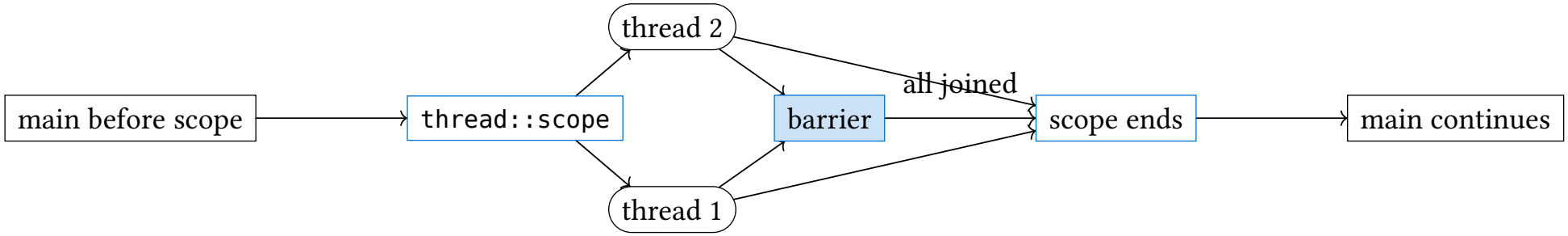

1.8. Moving to threads

In practice, scoped threads are not used very often.

The most common use case is to move ownership of data to the thread using the `move` keyword to transfer ownership at spawn time

```rust
fn main() {
    let data = String::from("hello");
    let handle = thread::spawn(move || {
        println!("Data: {data}"); // `data` moved into closure, now owned by thread
    });
    handle.join().unwrap();
}
```

(playground link)

Move can also be used for blocks (not closures):

1.9. Review: what is `'static`?

In the past sessions we have seen that `T: 'static` means:

Definition 1.9.1. *A type `T` is `'static` if it contains no non-`'static` references.*

Things that are `'static`:

- Owned data: `String`, `Vec<T>`, `Box<T>`, etc.
- `'static` references: `&'static str`, `&'static T`

Things that are not `'static`:

- References with shorter lifetimes: `&T`, `&'a T` where `'a` is not `'static`
- Types with non-`'static` fields: `&T`, `&'a T`, `Vec<&T>`, etc.

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Strings are owned and without references, so they are `'static` and can be moved

1.9.1. Spawn function

The standard library spawn function looks like this (simplified):

```
1 fn spawn<F, T>(f: F) -> JoinHandle<T>
2 where
3     F: FnOnce() -> T + 'static,
4     T: 'static                                    (playground link)
```

(Similar for asynchronous spawns like `tokio::spawn`, see next session.)

2.1. What is a channel?

A channel is a communication primitive for passing messages between threads.

Under the hood: a thread-safe buffer (queue) in shared memory with:

- **Sender** (`tx`): pushes messages into the buffer
- **Receiver** (`rx`): pulls messages from the buffer

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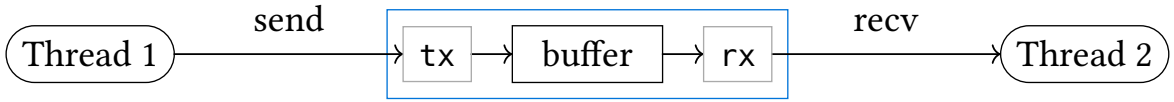

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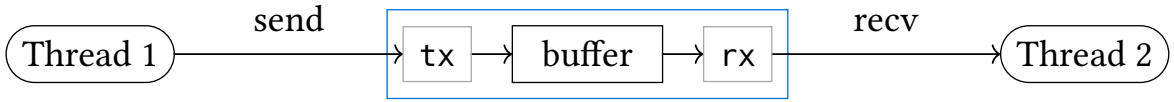

Key properties:

- **Decouples** sender and receiver (no shared mutable state)
- **Asynchronous**: sender doesn't wait for receiver (unless bounded)
- **Ownership transfer**: messages are moved, not shared

```rust
fn main() {
    let (tx, rx) = std::sync::mpsc::channel();
    std::thread::spawn(move || {
        let thread_id = std::thread::current().id();
        for i in 0..10 {
            tx.send(format!("Message {i}")).unwrap();
            println!("{thread_id:?}: sent Message {i}");
        }
        println!("{thread_id:?}: done");
    });
    std::thread::sleep(std::time::Duration::from_millis(100));

    for msg in rx.iter() {
        println!("Main: got {msg}");
    }
}
```

(playground link)

In an unbounded channel:

- `send()` never blocks (a kind of "asynchronous" behavior)
- returns a `Result::Err` when all receivers have been dropped

Definition 2.2.1. *A channel is "closed" when all receivers have been dropped.*

2.3. Bounded channels

If you don't want the channel buffer to grow indefinitely and cause memory overflow, you can use a bounded channel:

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```rust
fn main() {
    let (tx, rx) = mpsc::sync_channel(3);

    thread::spawn(move || {
        let thread_id = thread::current().id();
        for i in 0..10 {
            tx.send(format!("Message {i}")).unwrap();
            println!("{thread_id:?}: sent Message {i}");
        }
        println!("{thread_id:?}: done");
    });
    thread::sleep(Duration::from_millis(100));

    for msg in rx.iter() {
        println!("Main: got {msg}");
    }
}
```

(playground link)

This channel provides an additional method `try_send()` that returns immediately with an error if the buffer is full.

2.4. Exercises

- `examples/channel.rs`

3.1. Marker Traits

What are auto traits?

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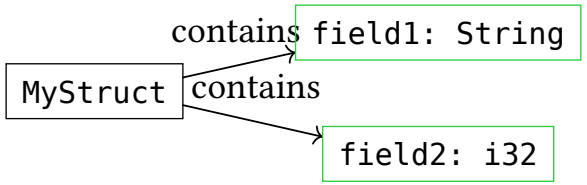
The compiler will automatically derive them for your types as long as they only contain types that implement the auto trait.



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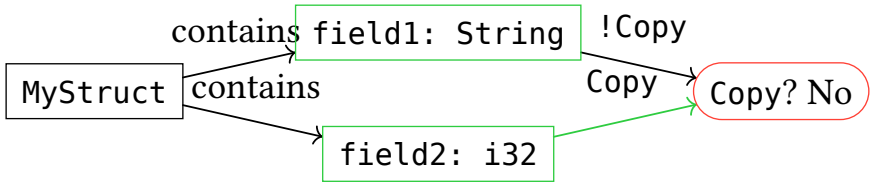

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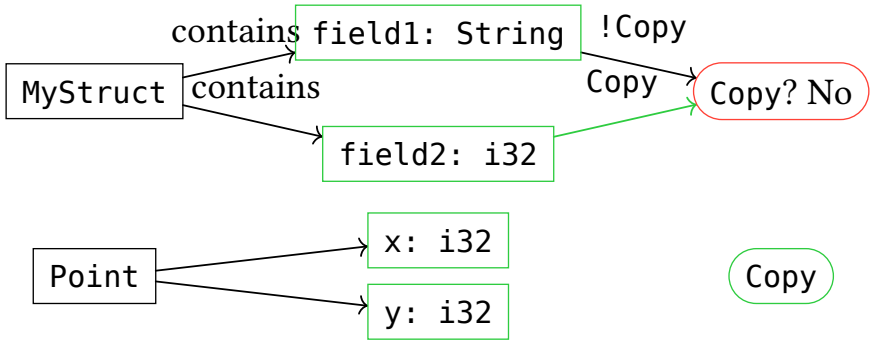

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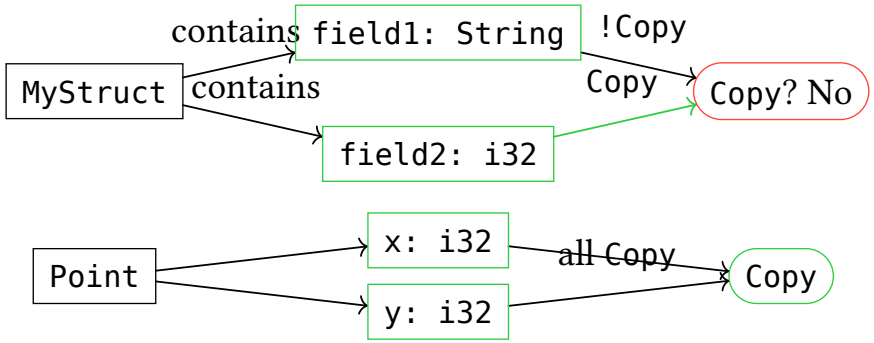

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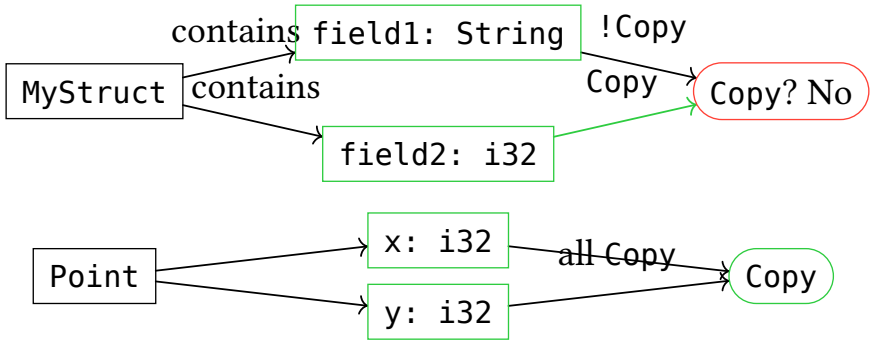

What are `unsafe` traits?

Traits that have safety invariants that the compiler cannot verify automatically. Implementing them incorrectly can lead to undefined behavior.

You can implement `unsafe` traits manually when you know it is valid.

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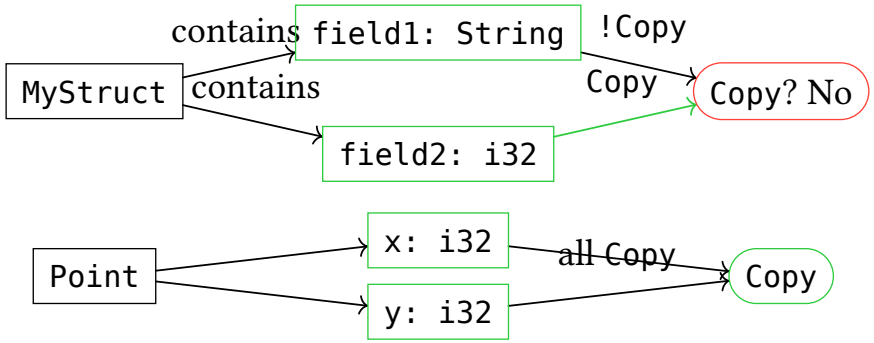

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You can implement `unsafe` traits manually when you know it is valid.

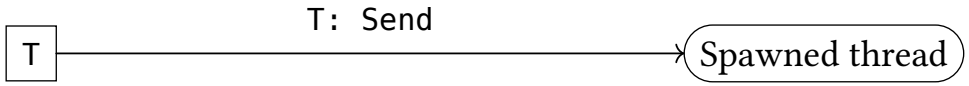
Warning

Implementing `unsafe` traits requires `unsafe` blocks. Using them as constraints does not.

3.2. Send and Sync

Definition 3.2.1.

- *`Send`: safe to **move** `T` to another thread*
- *`Sync`: safe to **share** `&T` across threads (i.e., `&T: Send`)*



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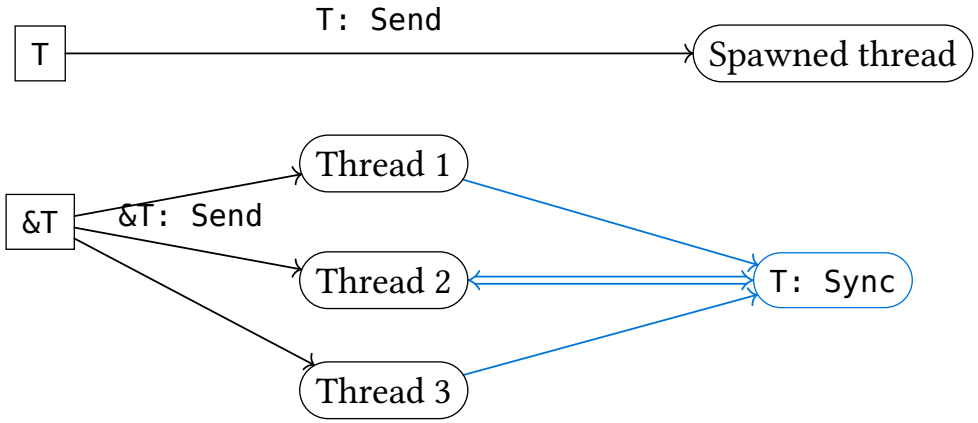

Info

Send and Sync are unsafe auto traits.

3.3. Send + Sync

Most types you come across are Send + Sync:

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When the type parameters `T` are Send + Sync.

3.4. Atomics

Atomics are low-level types for lock-free concurrent programming.

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Two problems with normal memory operations:

- Compilers and CPUs may **reorder** instructions for optimization
- Read-modify-write (e.g., `x += 1`) is **not atomic** (multiple CPU instructions)

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Two problems with normal memory operations:

- Compilers and CPUs may **reorder** instructions for optimization
- Read-modify-write (e.g., `x += 1`) is **not atomic** (multiple CPU instructions)

```rust
static mut DATA: u32 = 0;
static mut READY: bool = false;

fn thread1() { unsafe { DATA = 42; READY = true; } }
fn thread2() { unsafe { while !READY {} println!("{}", DATA); } } // might print 0!
```

(playground link)

CPU might execute `READY = true` before `DATA = 42`.

3.5. Atomics (continued)

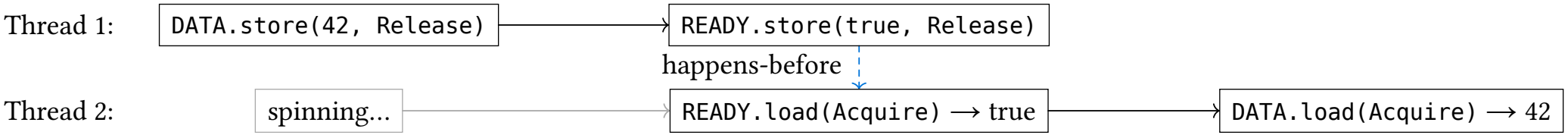
Solution: atomic operations with memory ordering guarantees.

```rust
use std::sync::atomic::{AtomicBool, AtomicU32, Ordering};

static DATA: AtomicU32 = AtomicU32::new(0);
static READY: AtomicBool = AtomicBool::new(false);

fn thread1() {
    DATA.store(42, Ordering::Release);
    READY.store(true, Ordering::Release);
}
fn thread2() {
    while !READY.load(Ordering::Acquire) {}
    println!("{}", DATA.load(Ordering::Acquire)); // guaranteed 42
}
```

(playground link)



The happens-before edge is on `READY` (same atomic, Release-store ⟶ Acquire-load).

This transitively makes `DATA.store` visible to `DATA.load`.

3.6. Atomics: hardware support

Not all CPUs support all atomic operations natively.

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- **Always supported**: `AtomicBool`, `AtomicU8`, `AtomicU16`, `AtomicU32`, `AtomicUsize`
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Check at compile time:

```
1 #[cfg(target_has_atomic = "64")]
2 use std::sync::atomic::AtomicU64;                    (playground link)
```

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Check at compile time:

```rust
#[cfg(target_has_atomic = "64")]
use std::sync::atomic::AtomicU64;
```

(playground link)

If hardware doesn't support an atomic size, Rust either:

- Won't compile (type not available)
- Falls back to a lock-based implementation (slower)

Info

On modern x86-64 and ARM64, all common atomics are hardware-supported.

O'REILLY®

Rust Atomics and Locks

Low-Level Concurrency in Practice

Mara Bos

3.7. Send + !Sync

These types can be moved to other threads, but cannot be shared via `&T`:

3.7. Send + !Sync

These types can be moved to other threads, but cannot be shared via `&T`:

- **Non-thread-safe interior mutability**: `Cell<T>`, `RefCell<T>`
  - Allow mutation through `&T` without synchronization
  - Safe to own exclusively on one thread, unsafe to share references

3.8. !Send + Sync

These types are safe to access (via shared references) from multiple threads, but cannot be moved to another thread:

- **Lock guards**: `MutexGuard`, `RwLockReadGuard`, `RwLockWriteGuard`
  - Must be unlocked on the thread that acquired the lock (POSIX requirement)

Proposition 3.8.1. *`MutexGuard<T: Sync>: !Send + Sync`*

3.9. !Send + !Sync

These types are not thread-safe and cannot be moved to other threads:

3.9. !Send + !Sync

These types are not thread-safe and cannot be moved to other threads:

- dereferences and has non-atomic count: `Rc<T>`
- considered unsafe in general `*const T`, `*mut T`

3.10. Summary

	Sync	!Sync

Send	`i32`, `bool`, `String`, `Vec<T>`
	`Arc<T>`, `Mutex<T>`, `Atomic*`
	Safe to move and share
		`Cell<T>`, `RefCell<T>`
		Interior mutability: safe to move, not share

!Send	`MutexGuard`, `RwLock*Guard` *Thread-bound: safe to share, not move*	`Rc<T>`, `*const T`, `*mut T` *Not thread-safe at all*

4.1. Sharing data

The unsafe way to share data between threads is to use `static mut`:

```
1 static mut COUNTER: u32 = 0;
2 fn increment() {
3     unsafe {
4         COUNTER += 1;
5     }
6 }
```

(playground link)

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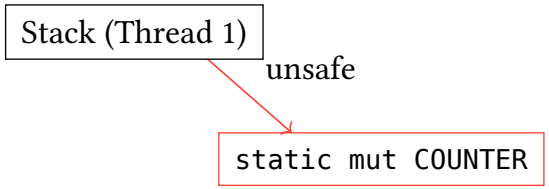

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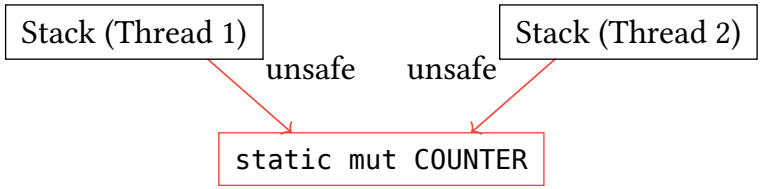

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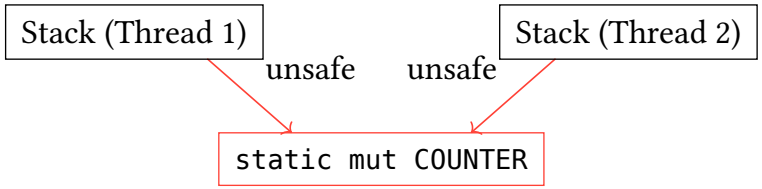

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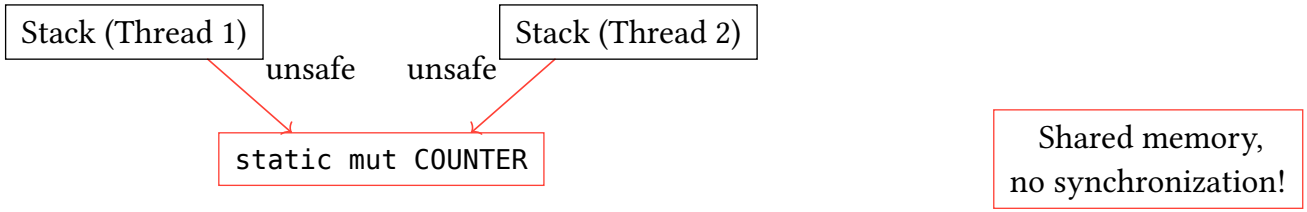

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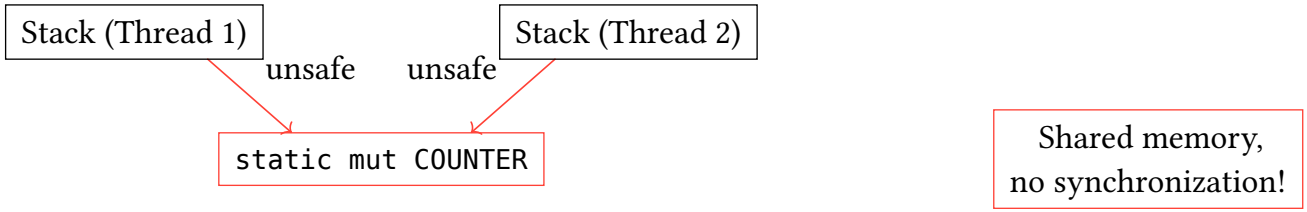

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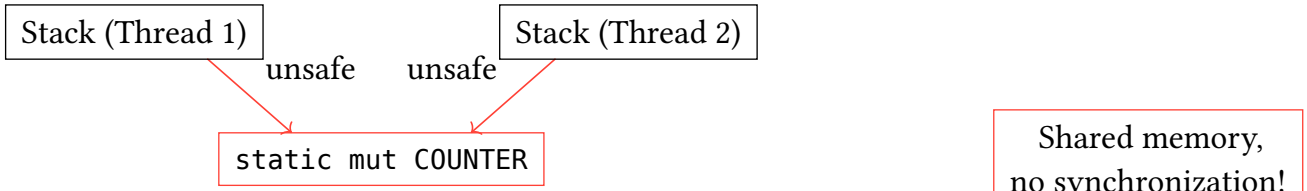

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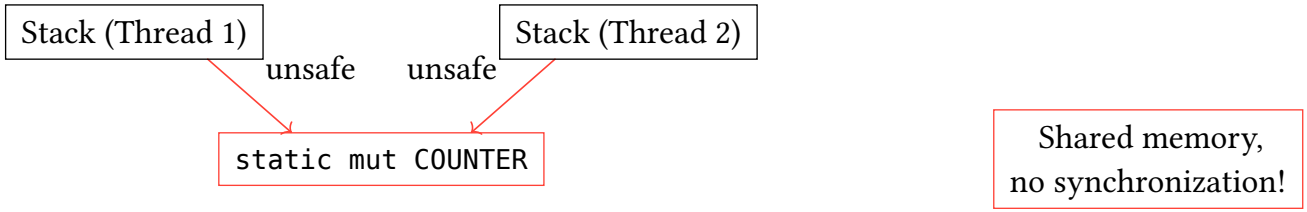


Disadvantages:

- Unsafe: requires `unsafe` blocks to access
- Needs to be initialized for the entire program lifetime

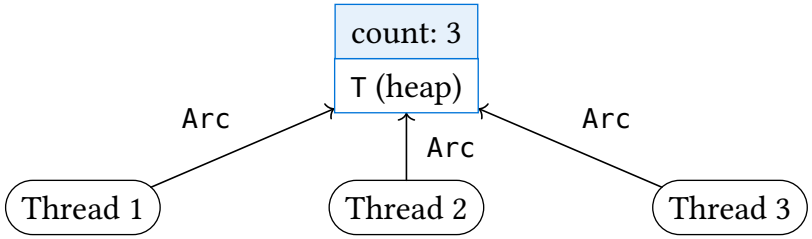
Solution:

1. Use reference counted variable (next slides)
2. Use a lock to ensure exclusive access (later)

(Notice that in this simple case we could just use an `AtomicU32` instead. This is just an example.)

4.2. Concept

Definition 4.2.1. `Arc<T>` = ***Atomic Reference Counted*** *pointer for shared ownership across threads.*



Key properties:

- `Arc::clone(&v)` increments ref count (atomic operation)
- When count reaches 0, data is dropped
- `Arc<T>: Send + Sync` when `T: Send + Sync`

4.2. Concept

4.2.1. Arc example

```rust
use std::sync::Arc;
use std::thread;

fn main() {
    let data = Arc::new(vec![1, 2, 3]);

    let handles: Vec<_> = (0..3).map(|i| {
        let data = Arc::clone(&data);
        thread::spawn(move || {
            println!("Thread {i}: {:?}", data);
        })
    }).collect();

    for h in handles { h.join().unwrap(); }
}
```

(playground link)

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(playground link)

Who drops the data?

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(playground link)

Who drops the data?

The last thread to finish (last `Arc` clone to go out of scope).

Can we mutate `T` through `Arc<T>`?

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use std::sync::Arc;
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(playground link)

Who drops the data?

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Can we mutate `T` through `Arc<T>`?

No, `Arc` only provides shared access. Use `Arc<Mutex<T>>` for mutation.

5.1. Mutex

Prevents race conditions on complex shared data using synchronisation.

How are you used to do data synchronisation in your favourite language?

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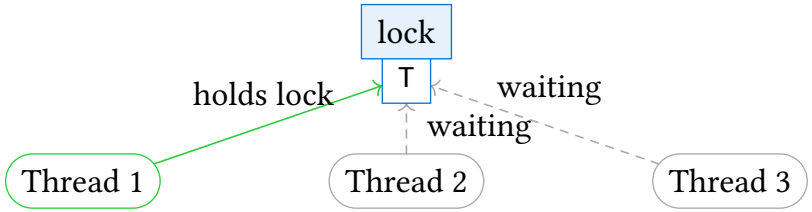
...

Definition 5.1.1. `Mutex<T>` = ***Mutual Exclusion*** *lock for exclusive access to* `T` *across threads.*



Key operations:

- `lock()` ⟶ blocks until lock acquired, returns `MutexGuard<T>`
- `MutexGuard` auto-releases lock when dropped (RAII)
- `Mutex<T>: Sync` when `T: Send`

5.1. Mutex

5.1.1. Mutex example

```rust
use std::sync::Mutex;

fn main() {
    let counter = Mutex::new(0);

    {
        let mut guard = counter.lock().unwrap();
        *guard += 1;
    } // lock released here

    println!("Counter: {:?}", counter.lock().unwrap());
}
```

(playground link)

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What is a `PoisonError`?

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What is a `PoisonError`?

Another thread panicked while holding the lock. The data may be in an inconsistent state.

Why does `Mutex<T>` require `T: Send` but not `T: Sync`?

5.1. Mutex

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(playground link)

What is a `PoisonError`?

Another thread panicked while holding the lock. The data may be in an inconsistent state.

Why does `Mutex<T>` require `T: Send` but not `T: Sync`?

The mutex guarantees only one thread accesses `T` at a time, so `T` doesn't need to support concurrent access.

5.1. Mutex

5.1.2. Arc + Mutex: shared mutable state

```rust
use std::sync::{Arc, Mutex};
use std::thread;

fn main() {
    let counter = Arc::new(Mutex::new(0));

    let handles: Vec<_> = (0..4).map(|_| {
        let counter = Arc::clone(&counter);
        thread::spawn(move || {
            let mut guard = counter.lock().unwrap();
            *guard += 1;
        })
    }).collect();

    for h in handles { h.join().unwrap(); }
    println!("Counter: {}", counter.lock().unwrap()); // 4
}
```

(playground link)

- `Arc` provides shared ownership across threads
- `Mutex` provides exclusive mutable access

5.2. Interior mutability

Definition 5.2.1 (Interior mutability). *A type that allows mutation through a shared reference (`&T` → `&mut` inner).*

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Types with interior mutability:

Type	Thread-safe?	Checking	Use case
`Cell<T>`	No	Compile-time	Simple values, `Copy` types
`RefCell<T>`	No	Runtime (panics)	Complex borrows, single-threaded
`Mutex<T>`	Yes	Runtime (blocks)	Exclusive access across threads
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`Atomic*`	Yes	Hardware	Counters, flags, lock-free

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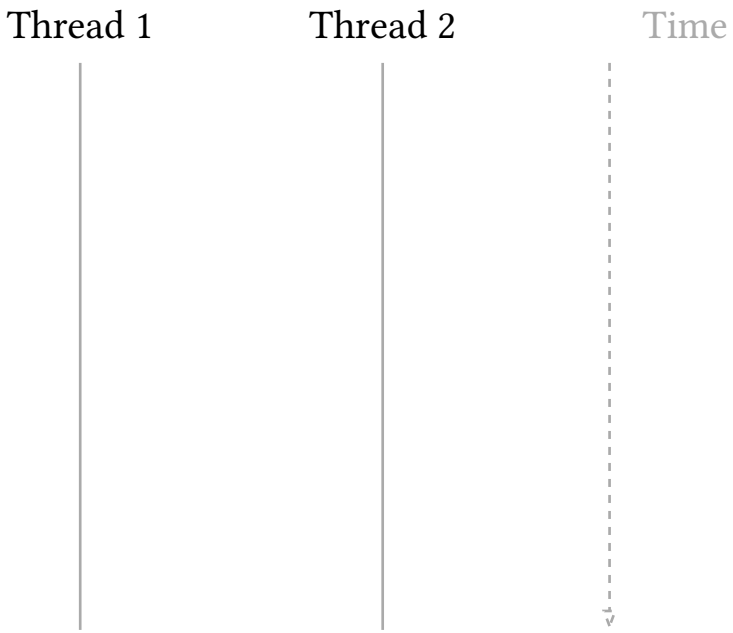
5.3. Deadlocks

The `Arc<Mutex<T>>` pattern is common but has drawbacks: forgetting to drop the guard before blocking operations



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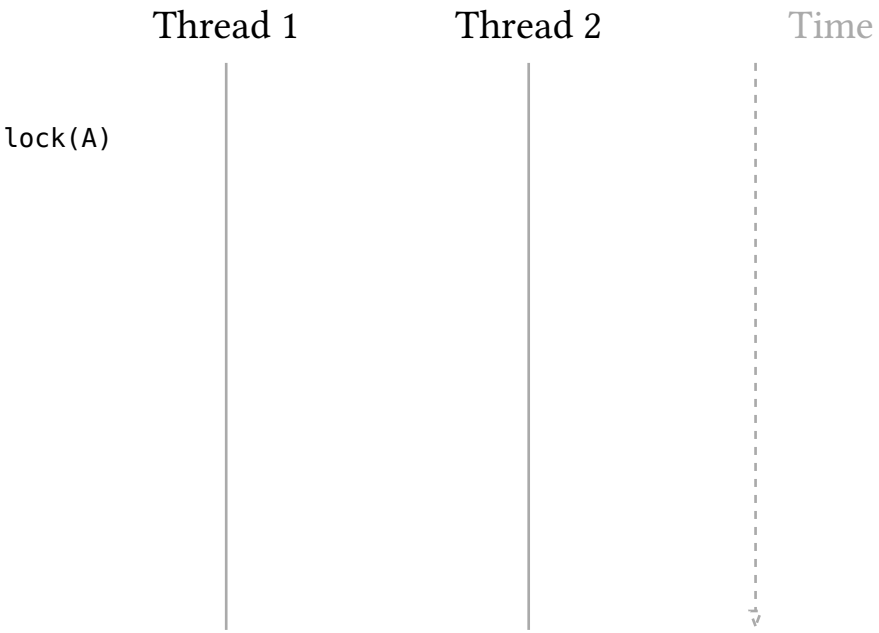

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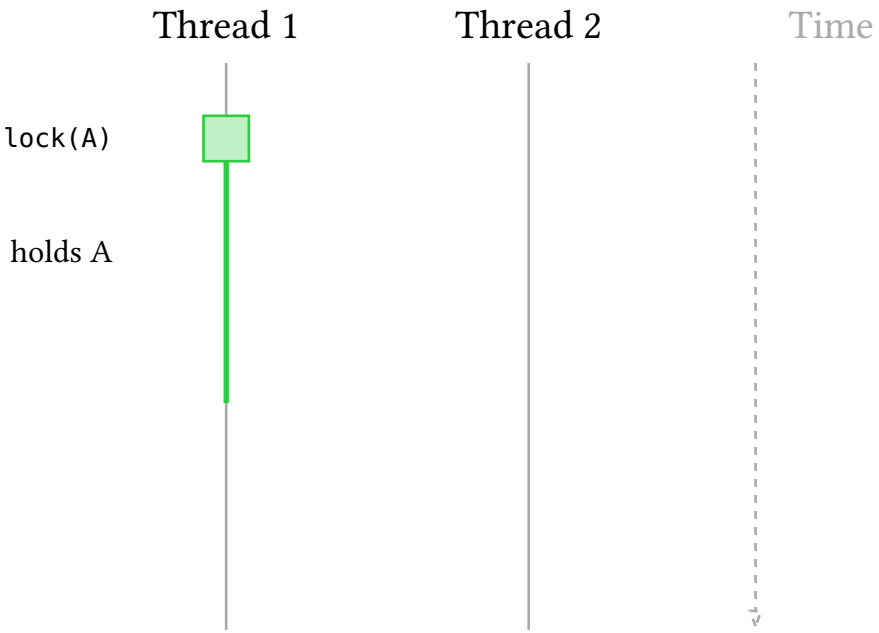

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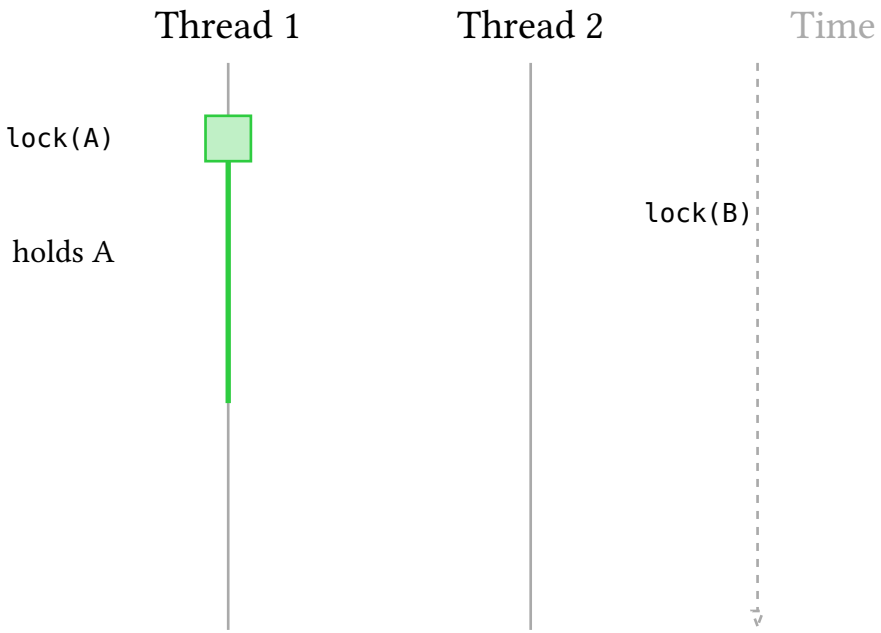

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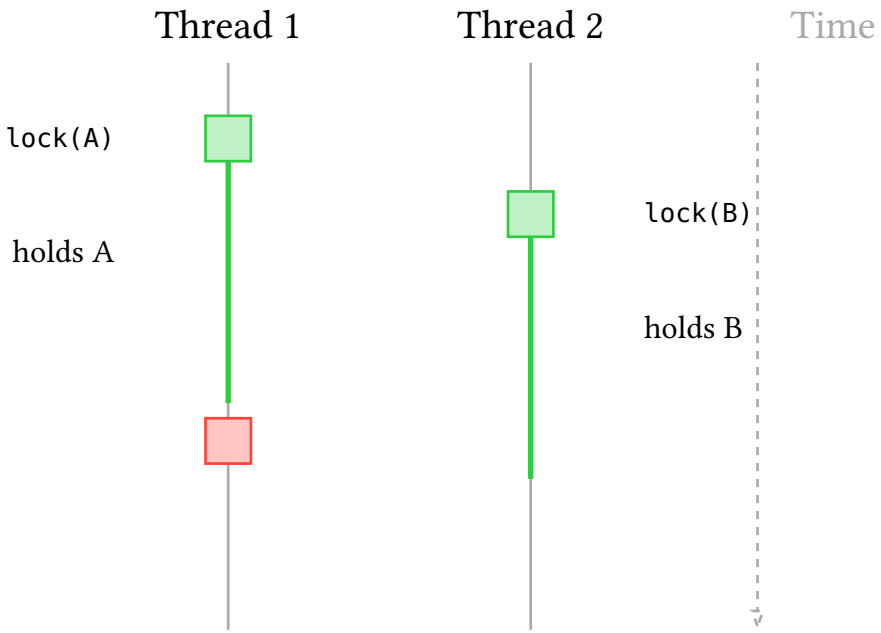

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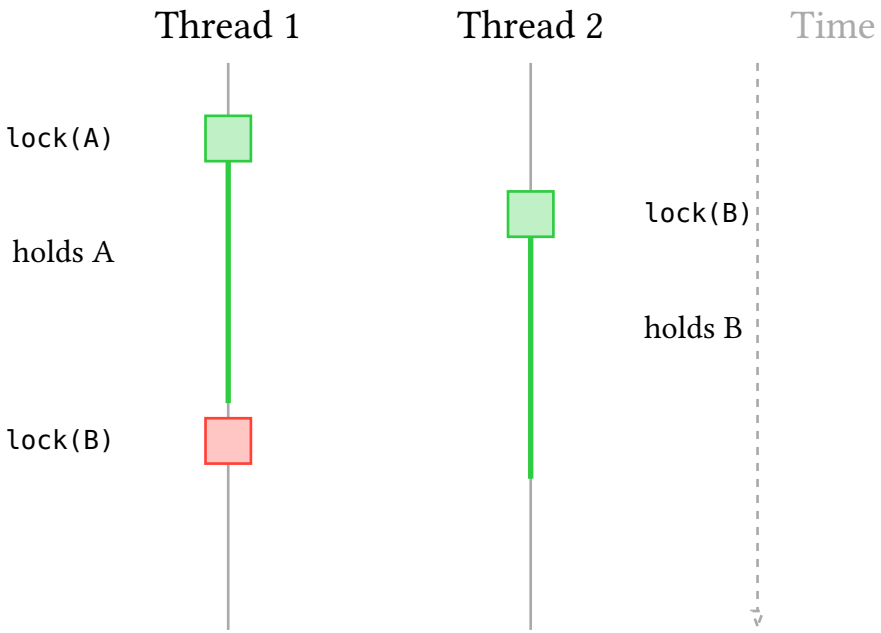

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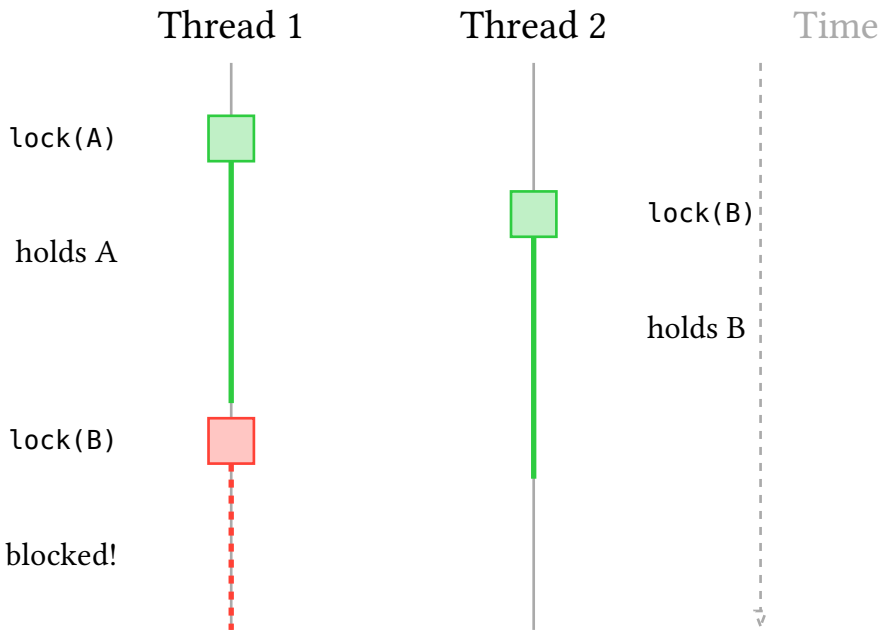

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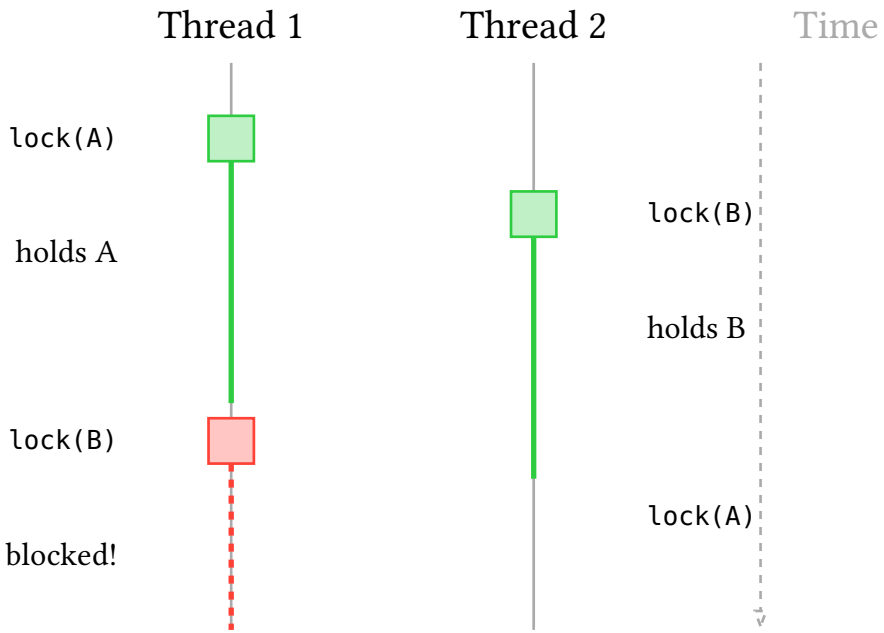

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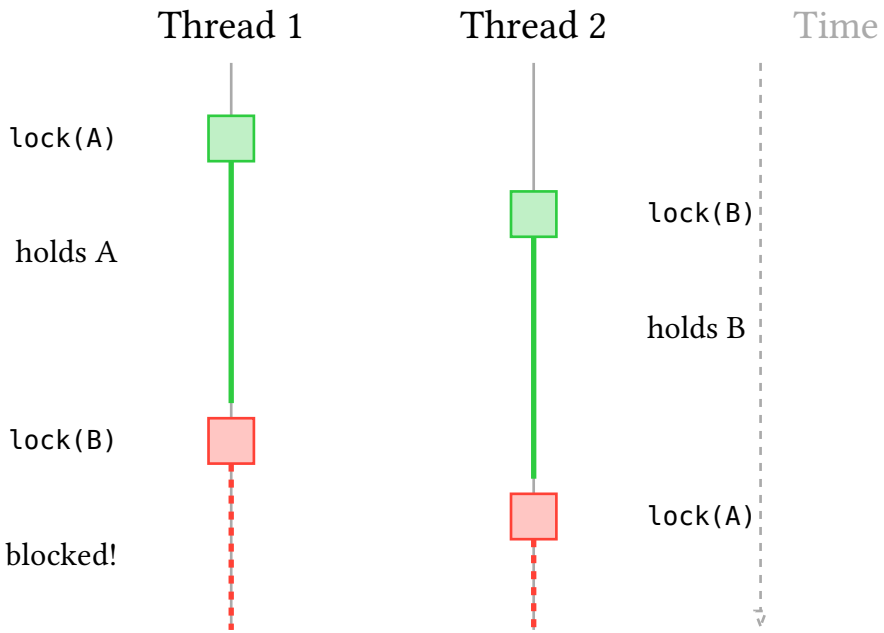

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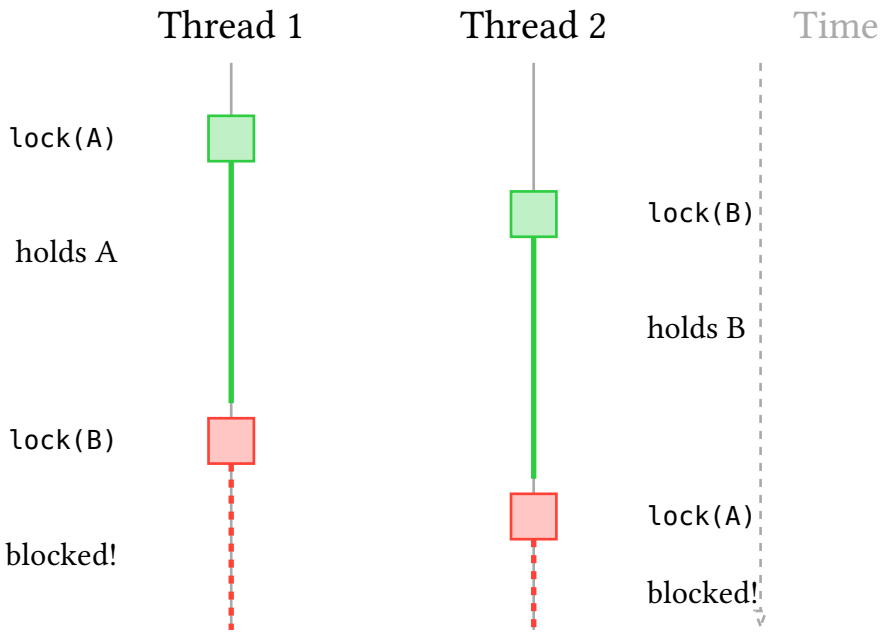

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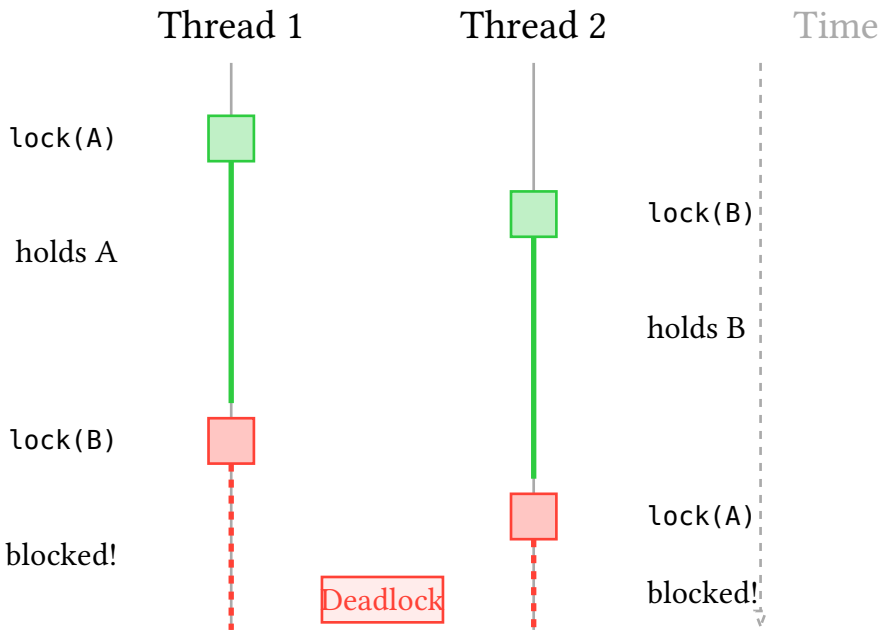

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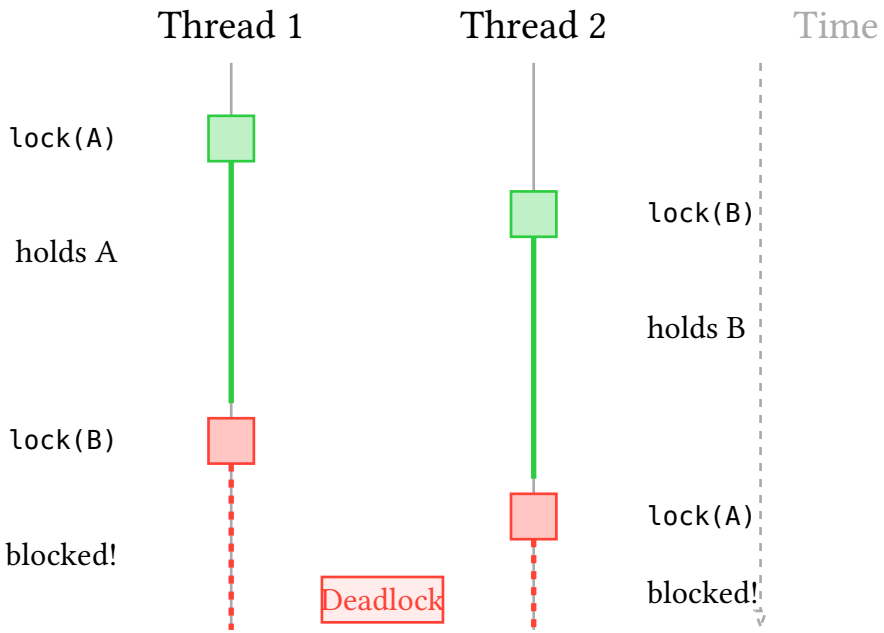

Exercise:

- examples/philosophers.rs

Solutions are provided as `*-solution.rs` files.

5.4. Best practices for Mutexes

Structure code to minimize shared mutable state in mutexes.

Alternatives:

- Use `RwLock<T>` for many readers, few writers
- Use `Atomic*` types for simple counters/flags

If really necessary:

- Use channels for **one-directional flow** (but they are also hard to maintain)

One more exercise:

- `examples/link.rs` (you need to have openssl installed, use the `flake.nix` or your package manager)

Solutions are provided as `*-solution.rs` files.